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When are urban growth boundaries not second-best policies to congestion tolls? ☆

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Abstract

Pines and Sadka proved that *a not-too-stringent urban growth boundary is a second-best policy to congestion tolls when traffic congestion is unpriced*, by assuming that all jobs are exogenously located at one urban center (monocentric city) [D. Pines, E. Sadka, Zoning, first-best, second-best and third-best criteria for allocating land to roads, Journal of Urban Economics 17 (1985) 167–183]. The result is also implied by Kanemoto [Y. Kanemoto, Cost-benefit analysis and the second-best land use for transportation, Journal of Urban Economics 4 (1977) 483–503] and Arnott [R. Arnott, Unpriced transport congestion, Journal of Economic Theory 21 (1979) 294–316]. Brueckner extrapolated this narrow theoretical result to real cities [J. Brueckner, Urban sprawl: Diagnosis and remedies, International Regional Science Review 23 (2000) 160–179]. We show that if there is no cross-commuting between city and suburb, first-best efficient tolls on traffic can reduce congestion and total travel cost by shifting worker-residents from the city to the suburbs, causing urban expansion. Then, planned urban boundaries of any stringency are not a second-best policy because they induce people to relocate to more congested areas. With cross-commuting, boundaries of any stringency can be inefficient even when tolls shrink cities, as boundaries do little but tolls do a lot to reduce inefficient suburb-to-city commuting. We also show that when the urban radius is limited by a natural boundary, then growth boundaries of any stringency are inefficient.
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Plato is dear to me, but dearer still is truth.
—Aristotle

1. Introduction

The concern with urban sprawl in recent years has led economists to grope for a proper framework to analyze the issue, and the effects of proposed remedies. Anas, Arnott and Small [1] have expressed views that emphasized the importance of models of employment dispersion, mixed jobs and residences and discretionary travel in explaining the structure of modern cities. Recently, Nechyba and Walsh [10] struck a similar note stressing the importance of sorting among heterogeneous communities.

These views do not find resonance in the standard monocentric model of urban areas. But Anas and Xu [3] is a general equilibrium model suited to modern cities that reflects the above mentioned views. The model deals with the endogenous location of production with flexible demand for land and labor inputs. Production and residences are endogenously mixed in all locations and sorting of consumer-workers among communities recognizes taste dispersion. Travel includes commuting as well as non-work dispersed discretionary trips and labor–leisure trade-offs. In contrast, in the monocentric model, the standard but anachronistic tool of urban economics, all jobs are pinned in the CBD by assumption. Hence, mixing of residences and production is ruled out, there is no non-work travel and taste dispersion is not recognized.

Using the Anas–Xu [3] model, we showed in Anas and Rhee [2] that an urban growth boundary (UGB) designed to contain urban sprawl in metropolitan areas with unpriced congestion is a blunt instrument with seriously negative welfare effects. In that article, excess sprawl was defined as excess time inefficiently spent in commuting and discretionary travel. To get the same travel reduction that can be achieved via first-best congestion tolls, the UGB would have to be extremely stringent, raising rents significantly, especially in the metropolitan edge communities and distorting wages and other prices, creating large deadweight losses. We also showed that congestion tolls can reduce sprawl significantly in the long run, but have small welfare benefits. In the simulations we reported, an urban boundary is not a second-best policy to congestion tolls. The boundary results in welfare losses no matter how slightly it is set.

This result, based on assumptions that recognize key facts about modern cities, can be contrasted with a set of mutually confirming theoretical exercises by Kanemoto [8], Arnott [4] and Pines and Sadka [11]. The last authors showed that in a city in which all commuting goes to jobs located at a single point or to a small Central Business District (CBD), and in which traffic congestion is unpriced, an urban boundary restricting the radius of the monocentric city is a second-best policy to congestion tolls. Although this second-best nature of the UGB was derived by Pines and Sadka in the context of a monocentric city,[1] Brueckner [5] claimed it to be true generally for real cities, while in a recent paper [6], written in response to the non-monocentric analysis of Anas and Rhee [2], he presented numerical simulations only for a standard monocentric city, confirming the result of Pines and Sadka [11].

Perhaps, the intuition behind Pines and Sadka's [11] result that the UGB is a second best policy to congestion tolls needs explaining here. As is well known, congestion tolls eliminate

[1] While Kanemoto [8] and Arnott [4] did not explicitly derive the second-best optimality of a UGB, this is implied by their analyses which show that the shadow rent on land at the fringe of the monocentric city with unpriced congestion is below the exogenous rent on agricultural land.

the pricing inefficiency because they are a Pigouvian tax on the negative externality of traffic congestion. By raising the cost of travel per mile to equal its social marginal cost on every mile, congestion tolls induce commuters to relocate their residences closer to the CBD and save on travel by economizing on distance to work which is their only margin of adjustment in a monocentric city.[2] As this happens, densities go up and more so near the center, total travel time to the CBD is reduced and, inevitably, the city shrinks. In the tolled city, the urban residential rent at the edge of the city equals the exogenous agricultural land rent as is also true in the un-tolled market city which is more sprawled. It follows then, that a planner could restrict up to an optimal extent, the radius of the market monocentric city with un-tolled congestion and get a second-best welfare improvement. The second-best welfare improvement happens because the UGB raises rents within the developed area by creating an artificial land scarcity. Faced with higher land prices, consumers reduce lot sizes and densities increase. This cuts average distance and travel time to the CBD reducing overall congestion. Thus, in the standard monocentric city, the direction of the effect of the restrictive boundary policy is the same as the direction of the effect of congestion tolls and that is why the UGB works as a second-best policy.

How is the disagreement between the standard monocentric models and the modern, non-monocentric general equilibrium analysis of Anas and Rhee [2] to be explained?

The answer lies in the fact that Anas and Rhee [2] differs in several important respects from the standard model by being a more complete and accurate theory of reality. In this article we use a simplified version of our model to illustrate why the UGB is often not a second-best policy to congestion tolls. In this simplified version, we suppress non-work travel and impose only two exogenously located job centers (one in the center of the city, the other in the suburbs) but we maintain the endogenous choice of workplace (between the two centers) and of place of residence (between city and suburb), and we also maintain the dispersion of tastes.

The first feature we explore in Sections 3 and 4 is that in our model, as opposed to the standard monocentric model of urban economics, jobs as well as residences are mobile and commuters travel in all directions not just to a central point. Table 1 shows the commuting patterns in US and Canadian cities as recently documented by the respective censuses. Imposing congestion tolls or an urban boundary will cause both jobs and residences to shift locations within the city. In particular, jobs are not pinned in the CBD as monocentric analysis arbitrarily assumes and can be reallocated to the suburban job centers.

In the simulations reported in Section 3, we assume that jobs and residences are linked but that consumer-workers are free to make the binary choice to reside-work in the city or to reside-work in the suburbs. Thus all commuting is local and happens either in the city or in the suburbs.

Table 1
Commuting patterns in USA and Canada

Commuting patterns		United States	Canada
Residence	Workplace	2000 Census (%)	2001 Census (%)
Central city	Central city	27.5	46.1
Central city	Suburb	8.9	7.5
Suburb	Central city	20.2	16.2
Suburb	Suburb	43.4	30.2
		100.0	100.0

[2] Unlike Anas and Rhee [2], the monocentric models consider only work travel and ignore idiosyncratic tastes.

There is no cross-commuting. Using this setup, our numerical solutions are presented in Fig. 3 which shows examples, tabulated in Tables 2 and 3. In the examples of Table 2, a UGB of limited stringency which is a second-best policy to congestion tolls does not exist. The common feature of all these cases is that congestion tolls cause residents and their jobs to be relocated to the suburbs expanding the urban area. In these cases, it is implied that, when tolling of congestion is not possible, then the planner should zone the urban area for expansion at the expense of agriculture rather than the other way around as conventional urban economics presumes. The examples in which a UGB of limited stringency is a second-best policy to congestion tolls are tabulated in Table 3. The common feature of these examples is that congestion tolls cause residents and their jobs to move from the suburbs to the city shrinking the urban area, so the conventional thinking, derived from monocentric models, about UGBs that restrict urban expansion as second-best policies to congestion tolls is valid.

The simulations of Section 4 are more general since there we allow consumer-workers to choose among all four possible commuting arrangements that are empirically documented in Table 1: reside in city, work in suburbs; reside in suburbs, work in city; reside and work in city; and reside and work in suburbs. In this context, imposing tolls could make travel in and to the central city too expensive, causing some to switch their jobs to a Suburban Business District (SBD) and others to move their residence closer to their CBD jobs. Still others could make both switches simultaneously, flipping their residences from the suburbs to the city and their jobs from the CBD to the SBD, reverse commuting from city to suburbs. In our model, the relative frequency of these adjustments is explained by the distribution of idiosyncratic tastes in the population. The results are shown in Fig. 4 and tabulated in Table 4. The table includes an example in which the restrictive UGB is a second-best policy and two examples in which it is not.

In the simulations of Section 3, with cross-commuting ruled out, the UGB is second-best efficient whenever tolls cause the urban area to shrink. But when tolls cause the urban area to expand by causing worker-residents to move to the suburbs, then the UGB is not second-best efficient. In the simulations of Section 4, where cross-commuting is allowed, similar results are obtained. However, in these simulations, the UGB can be inefficient even when tolls shrink the urban area. The reason for this is that the UGB, unlike tolls, does little to discourage suburb-to-city commuting but instead discourages suburb-to-suburb commuting and causes suburban road capacity to be underutilized.

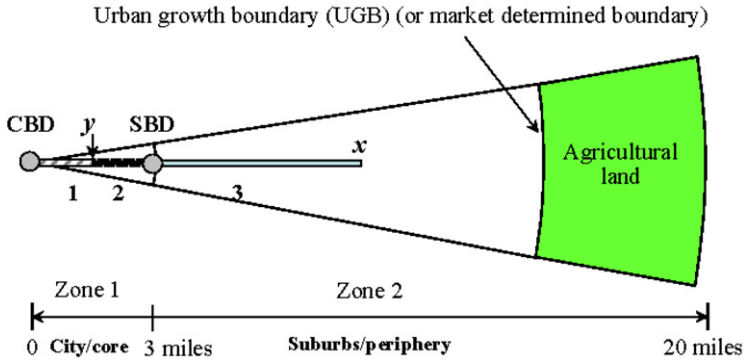
The second feature we explore in Section 5 relates to the assumption of a high exogenous agricultural rent that limits the extent of a market city. Although this assumption is almost universal in the standard monocentric models, it often fails to be true. For one thing, there are many urban areas that have expanded to a natural boundary. Los Angeles surrounded by mountains is a possible example. Las Vegas, located in the middle of the desert, is surrounded by land that has little or no value, but a "natural" border exists because land beyond a certain distance is government owned and has been kept out of development. But even in those cases where surrounding land is in agriculture, it is not reasonable to assume that its rent is exogenous. Muth [9] showed, in a static model, that as land within the city becomes more valuable so does the surrounding agricultural land. The premium over and above the rent in farming is known as option value. Dynamic models that treat the growth premium on surrounding land do exist (e.g. Capozza and Helsley [7]), but have not been used extensively by urban economists who still prefer the standard and static monocentric model. But if the purpose of using the static model is to gain insights about the reality, then an exogenous agricultural rent is not a good assumption. For that reason, in Section 5, we relax this assumption of an exogenous agricultural land rent. We simply assume

that there is no alternative land use for the land surrounding the urban area. In that case the urban area extends to occupy all of the available land up to the natural limit and suburban land rent becomes endogenous. Our numerical examples, presented in Fig. 5 and Table 5, show that, in such a setting, a boundary of any stringency is not a second best.

2. A simple city-suburb model

2.1. The setting

Figure 1 shows the simple urban setting we will examine. There are two zones representing a pie slice of a complete circular city. The slice corresponds to a transport corridor around a radially oriented sequence of roads and circumferential travel is ignored. Zone 1 has a 3-mile radius. This represents the central and most congested area of cities. At equilibrium, land use densities will be high there. Zone 2 represents the rest of the urban area (mostly low density central city and suburbs and surrounding agricultural area) and has a radial length of 17 miles. Jobs can be located at two predetermined centers: the CBD, located at the center of the city, and the Suburban Business District (SBD), located at the border between the two zones. It is assumed that the two work centers do not take up land. A road extends from point x in zone 2, to the SBD, extends on to point y in the center city and on to the CBD. Points x and y are "ramps" where commuters get on or off the road. Zone 1 residents (dis)embark at y and suburban residents at x. The cost of intra-zonal travel to access x or y is ignored. Only travel on the road is costly and those using the road can travel in either direction. Without loss of generality or much loss of realism we assume that traffic moving in opposite or the same direction on a road, congest that road equally. As shown in Fig. 1, there are three roads (road 1: y to/from CBD; road 2: SBD to/from y; road 3: x to/from SBD). The capacity of each road is fixed and each road can have a different capacity. This setup replicates all of the basic conditions assumed by Kanemoto, Arnott, Pines and Sadka but uses the simplest possible discrete version.[3] Differences from these earlier setups come about by replacing the unrealistic assumptions of monocentricity with the duocentric structure of a CBD/SBD and by replacing the strict inward commuting property of the monocentric



Road 1: y to from *CBD*; **Road 2**: *SBD* to from y; **Road 3**: x to from *SBD*.

Fig. 1. Unit slice of circular city with CBD and SBD.

[3] To be accurate, the capacity of roads in Kanemoto [8] is not fixed, but this difference is not a crucial one for the issues examined in this paper.

model by a structure of idiosyncratic taste dispersion that allows diverse commuting patterns such as those of Table 1.

Consumers can work either at the CBD or the SBD. Consumers can also reside at either the city (zone 1) or the suburbs (zone 2). The lot size purchased by a consumer is endogenously determined. In zone 2, there is an exogenously given agricultural rent, R_a.When congestion tolls are or are not levied, zone 2 will be divided into two parts by land arbitrage so that its outer part (shaded area in Fig. 1) will be in agriculture and the inner part in urban use. In both parts of zone 2, the equilibrium land rent will be equal to R_a. The amount of agricultural land area, A, will be determined by the arbitrage in land. Likewise, when congestion tolls are levied, an A will again be determined by land arbitrage. We will show that, $A_{\text{TOLLS}} \leqslant (\text{or } \geqslant) A_{\text{NO-TOLLS}}$. This result contrasts with the standard monocentric models where tolls shrink the urban area ($A_{\text{TOLLS}} > A_{\text{NO-TOLLS}}$). When the planner is assumed to set a UGB, arbitrage in the urban fringe is not possible. We assume that the planner will set a second-best welfare maximizing UGB in the absence of tolls such that $A^*_{\text{UGB}} \geqslant A_{\text{NO-TOLLS}}$. Then, when $A^*_{\text{UGB}} > A_{\text{NO-TOLLS}}$ in the equilibrium constrained by the planner's A, the rent on land in the inner (residential) part of zone 2 will be higher than the agricultural land rent in the outer part of zone 2: $R_2 > R_a$. Although we will not examine the case where $A^*_{\text{UGB}} < A_{\text{NO-TOLLS}}$, our analysis clearly implies that such a negative second-best UGB does exist when the effect of first-best congestion tolls is to expand the urban area.

2.2. *Preferences and choice probabilities*

The utility function of the consumer is $U = a \ln z + b \ln q$, with $a + b = 1$ and $0 < a, b < 1$. z is a numeraire composite good while q is the consumer's lot size. The consumer maximizes this utility with respect to the budget constraint $y - T = z + Rq$, where R is the rent on land, y is the consumer's income and T is the consumer's commuting cost. Suppose that the consumer has preselected his job and residence zone. Then, his income, his unit rent and his commuting cost are given to the consumer according to the choice of workplace, the choice of residence and the commuting arrangement. The two demand functions are:

$$z = a(y - T), \tag{1}$$

$$q = b\left(\frac{y - T}{R}\right). \tag{2}$$

The maximized (indirect) utility is

$$V = \ln(y - T) - b \ln R. \tag{3}$$

For a consumer who resides in zone i and works in (commutes to) zone j, we write his utility as $\tilde{V}_{ij} = V_{ij} + u_{ij}$, where $V_{ij} = \ln(y_j - T_{ij}) - b \ln R_i$ and the additive u_{ij} is an idiosyncratic constant that differs among consumers, measuring the pure individual preference for residing in zone i and working in zone j (i.e. for the commuting arrangement (i, j)). Each consumer has four such idiosyncratic constants, u_{11}, u_{12}, u_{21}, u_{22}. We assume that these are distributed i.i.d. over the population of consumers and are each drawn, for all consumers, from the Gumbel distribution with mean zero and variance $\sigma^2 = \frac{\pi}{6\lambda^2}$, where λ is called the "taste heterogeneity parameter." The probability that a randomly identified consumer will choose commuting arrangement i, j to maximize utility is then calculated as the following quatrinomial logit model:

$$P_{ij} = \text{Prob}\big(\tilde{V}_{ij} > \tilde{V}_{kl},\, \Delta_{ij} = 1,\ \forall (k,l) \neq (i,j) \text{ and } \Delta_{kl} = 1\big)$$
$$= \frac{\Delta_{ij} \exp[\lambda \ln(y_j - T_{ij}) - \lambda b \ln R_i]}{\sum_{k=1}^{2}\sum_{\ell=1}^{2} \Delta_{k\ell} \exp[\lambda \ln(y_\ell - T_{k\ell}) - \lambda b \ln R_k]}, \quad \left(\sum_{i=1}^{2}\sum_{j=1}^{2} P_{ij} = 1\right). \tag{4}$$

The Δ_{ij} are four zero-one constants created for our convenience, so that when $\Delta_{ij} = 0$, then $P_{ij} = 0$. We will set $\Delta_{ij} = 0$ if we do not allow the consumers to choose arrangement (i, j) and we will set $\Delta_{ij} = 1$ if we do. We will work either with $\Delta_{11} = \Delta_{22} = 1$ and $\Delta_{12} = \Delta_{21} = 0$ (cross-commuting is not allowed) or all $\Delta_{ij} = 1$ (cross-commuting is allowed).

The expected welfare measure is:

$$W = E\Big[\max_{\forall (i,j),\, \Delta_{ij}=1} \big(\tilde{V}_{ij}\big)\Big] = \frac{1}{\lambda} \ln \sum_{i=1}^{2}\sum_{j=1}^{2} \Delta_{ij} \exp(\lambda V_{ij}). \tag{5}$$

We will use this measure to evaluate the welfare gains from congestion tolling and any gains or losses from urban boundaries. A discrete choice model is a good way to capture the cross-commuting and dispersed location behavior identified in Table 1.

2.3. Congestion

T_{ij} is the total annual cost of commuting for a commuter who resides in zone i and works in zone j. Define T_s to be the two-way total cost required to traverse roads $s = 1, 2, 3$. Then, from the assumed road configuration of Fig. 1,

$$T_{11} = T_1, \tag{6a}$$
$$T_{22} = T_3, \tag{6b}$$
$$T_{21} = T_1 + T_2 + T_3, \tag{6c}$$
$$T_{12} = T_2. \tag{6d}$$

This total cost is the sum of the privately incurred congested travel costs, g_s, plus the congestion tolls, t_s, if any, on roads s traversed. Thus, $T_s = g_s + t_s$. The congested travel cost is given by the link congestion function

$$g_s = \alpha_s \left[1 + \beta \left(\frac{F_s}{\zeta K_s}\right)^{\gamma}\right], \tag{7}$$

where K_s is the link's capacity measured by the land area taken up by the road, and α_s, β, ζ, γ are positive constants used to calibrate the model. F_s is the flow of traffic on road s. Note that

$$F_1 = N(P_{11} + P_{21}), \tag{8a}$$
$$F_2 = N(P_{12} + P_{21}), \tag{8b}$$
$$F_3 = N(P_{21} + P_{22}), \tag{8c}$$

where N is the exogenously given total number of consumer-workers. α_s in (7) measures the un-congested (free flow) travel cost on road s. In reality, this is determined by a combination of the road's length, quality, speed limits, etc. If a first-best congestion toll is imposed on all roads s, then the toll is the difference between the social marginal and private average cost of F_s, which is

$$t_s \equiv \frac{\partial(g_s F_s)}{\partial F_s} - g_s = \gamma \alpha_s \beta \left(\frac{F_s}{\zeta K_s}\right)^{\gamma}. \tag{9}$$

2.4. Equilibrium conditions

We are now ready for the equilibrium conditions. There are three cases:

(1) *Market or laissez-faire*: Traffic is congested but there are no tolls on traffic,
(2) *Tolls*: Traffic is congested and first-best tolls are imposed on the traffic by a road authority, and
(3) *UGB*: Traffic is congested and there are no tolls but a restrictive UGB that maximizes welfare, if one exists, is imposed by land use planners to limit the urban radius in the urban–rural fringe of zone 2.

In each of the three cases the capacity of each road and the congestion functions stay fixed as is true also in Arnott [4] and Pines and Sadka [11].

Salient aspects of equilibrium in the two land markets are depicted by the supply–demand diagrams of Fig. 2. The demands to rent land in either zone are elastic and the cross-price elasticity is positive (the two zones are imperfect substitutes). In zone 1, the supply of land is perfectly inelastic. In zone 2, the supply of land is perfectly elastic until all of the available land is developed,

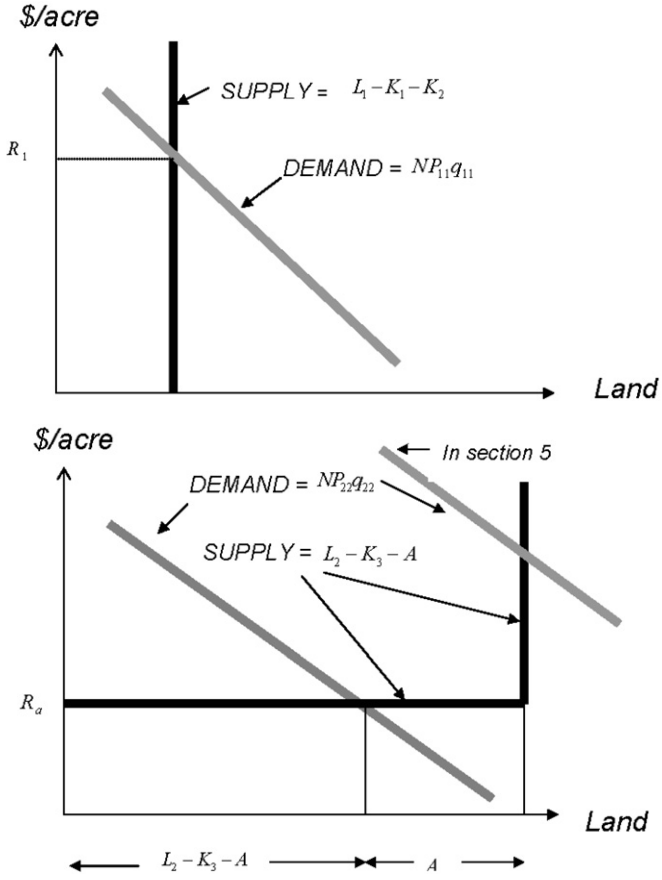


Fig. 2. Equilibrium in the two-zone model.

becoming perfectly inelastic thereafter. It similarly becomes perfectly inelastic at the margin if a binding UGB is introduced. The rent in the rural section of zone 2 will always be anchored at R_a, the exogenous agricultural rent. The rent in the urban part of zone 2 depends on whether a restrictive UGB is installed or not. When it is installed, the rent in the urban part of zone 2 will be endogenous and equal to $R_2 > R_a$. When a restrictive UGB is not installed, then the same R_a prevails by arbitrage in the urban and rural parts of zone 2, whether congestion tolls are imposed or not. The only exception to this occurs when there is no agricultural part in zone 2, namely the urban area extends all the way to the natural border occupying all of zone 2. In that case, the demand curve in zone 2 intersects the vertical part of the land supply curve and the endogenous rent in zone 2 under tolls will be different than under no-tolls and different still, if a restrictive UGB is installed.

Now consider the effect of congestion tolls in the context of Fig. 2. We will show later that tolls cause the two demand curves in Fig. 2 to shift up or down, but in opposite directions. If tolls cause the demand for land in zone 1 to shift up, rent and densities in zone 1 will increase. Meanwhile, the demand for land in zone 2 will shift down. Since the supply in zone 2 is perfectly elastic, the demand contraction will cause a reduction in the land developed and the urban radius will shrink. If, conversely, tolls cause the demand for land in zone 1 to shift down, rent in zone 1 will fall while the demand for land in zone 2 will increase, causing the urban radius to expand.

Market: *Unpriced congestion*

We develop the equilibrium equations for the general case, assuming that all commuting patterns are allowed (all $\Delta_{ij} = 1$.) First are two equations that clear the land market in each zone by equating the aggregate land demanded by the consumers to the land that is available after land taken up by roads and by farming is subtracted from the total land supply. These are:

$$NP_{11}b\left(\frac{y_1 - T_1}{R_1}\right) + NP_{12}b\left(\frac{y_2 - T_2}{R_1}\right) = L_1 - K_1 - K_2 \quad \text{(city land market)}, \tag{10}$$

$$NP_{21}b\left(\frac{y_1 - T_1 - T_2 - T_3}{R_a}\right) + NP_{22}b\left(\frac{y_2 - T_3}{R_a}\right) = L_2 - K_3 - A$$
$$\text{(suburban land market)}. \tag{11}$$

On the left sides of (10) and (11), NP_{ij} is the number of consumers choosing commuting pattern (i, j) and $q_{ij} = b(\frac{y_j - T_{ij}}{R_i})$ is the demand for lot size (given by (2)) of each consumer choosing (i, j). On the right sides, L_i is the total land available in zone i, K_i is the land taken up by road i and A is the agricultural land in zone 2. The reader should refer to Fig. 1.

Next are the road congestion equations:

$$T_1 = g_1 = \alpha_1\left(1 + \beta\left[\frac{NP_{11} + NP_{21}}{\zeta K_1}\right]^{\gamma}\right) \quad \text{(road 1 congestion function)}, \tag{12}$$

$$T_2 = g_2 = \alpha_2\left(1 + \beta\left[\frac{NP_{21} + NP_{12}}{\zeta K_2}\right]^{\gamma}\right) \quad \text{(road 2 congestion function)}, \tag{13}$$

$$T_3 = g_3 = \alpha_3\left(1 + \beta\left[\frac{NP_{21} + NP_{22}}{\zeta K_3}\right]^{\gamma}\right) \quad \text{(road 3 congestion function)}. \tag{14}$$

There are two more equations that show how the consumer's income is determined:

$$y_1 = y_{01} + (R_1 - R_a)\frac{L_1 - K_1 - K_2}{N} - \frac{R_a(\sum_{i=1}^{3} K_i)}{N}, \tag{15}$$

$$y_2 = y_{02} + (R_1 - R_a)\frac{L_1 - K_1 - K_2}{N} - \frac{R_a(\sum_{i=1}^{3} K_i)}{N}. \tag{16}$$

(15) and (16) say that income is composed of the exogenous wage received at the workplace (y_{01} or y_{02}) plus the aggregate net rent, hereafter briefly named D, distributed equally among all consumers. This dividend is the residential rent in excess of the agricultural rent collected from zone 1 less the cost of renting the land used for roads at its agricultural rent. Since in zone 2, the land rent is R_a there is no excess rent from zone 2. This is the standard assumption in urban economics and presumes that the urban land has been rented from the farmers (the initial owners) at the agricultural rent, and is again consistent with Kanemoto [8], Arnott [4] and the Pines and Sadka [11] models. We relax this assumption in Section 5.

The above seven equations (10)–(16) are to be solved for the endogenous variables R_1, A, T_1, T_2, T_3, y_1, y_2, given the exogenous variables R_a, L_1, L_2, K_1, K_2, K_3, y_{01}, y_{02} and N.

Tolls: *Priced congestion*

In this case, Eqs. (10) and (11) stay the same but (12)–(14) need to be modified by including the first-best toll in the travel cost, as explained earlier, and (15), (16) need to be modified to reflect that aggregate toll revenue is redistributed. (12)–(16) are modified as follows:

$$T_1 = g_1 + t_1 = \alpha_1\left(1 + \beta(\gamma+1)\left[\frac{NP_{11} + NP_{21}}{\zeta K_1}\right]^{\gamma}\right), \tag{17}$$

$$T_2 = g_2 + t_2 = \alpha_2\left(1 + \beta(\gamma+1)\left[\frac{NP_{21} + NP_{12}}{\zeta K_2}\right]^{\gamma}\right), \tag{18}$$

$$T_3 = g_3 + t_3 = \alpha_3\left(1 + \beta(\gamma+1)\left[\frac{NP_{21} + NP_{22}}{\zeta K_3}\right]^{\gamma}\right), \tag{19}$$

$$y_1 = y_{01} + (R_1 - R_a)\frac{L_1 - K_1 - K_2}{N} - \frac{R_a(\sum_{i=1}^{3} K_i)}{N} + \frac{TOLLS}{N}, \tag{20}$$

$$y_2 = y_{02} + (R_1 - R_a)\frac{L_1 - K_1 - K_2}{N} - \frac{R_a(\sum_{i=1}^{3} K_i)}{N} + \frac{TOLLS}{N}, \quad \text{where} \tag{21}$$

$$TOLLS \equiv \sum_{s=1}^{3} F_s t_s = \frac{\beta\gamma}{\zeta^{\gamma}}\left[\alpha_1\frac{(NP_{11} + NP_{21})^{\gamma+1}}{K_1^{\gamma}} + \alpha_2\frac{(NP_{12} + NP_{21})^{\gamma+1}}{K_2^{\gamma}} + \alpha_3\frac{(NP_{22} + NP_{21})^{\gamma+1}}{K_3^{\gamma}}\right].$$

In this case, the seven equations (10), (11) and (17)–(21) are to be solved for the endogenous variables R_1, A, T_1, T_2, T_3, y_1, y_2, given the exogenous variables R_a, L_1, L_2, K_1, K_2, K_3, y_{01}, y_{02} and N. Compared to the case of an equilibrium without tolls, the endogenous and exogenous variables are the same, but tolling will cause their values to differ.

Unpriced congestion with UGB

The following modifications are needed to the equations of the unpriced congestion case. R_a is replaced by R_2 in the suburban land market equation and the rent redistribution equations change because $R_2 \geqslant R_a$ in the last zone and there is now differential land rent revenue not only from zone 1, but also from zone 2, after the imposition of the restrictive UGB:

$$NP_{21}b\left(\frac{y_1 - T_1 - T_2 - T_3}{R_2}\right) + NP_{22}b\left(\frac{y_2 - T_3}{R_2}\right) = L_2 - K_3 - A$$

(suburban land market), (22)

$$y_1 = y_{01} + (R_1 - R_a)\frac{L_1 - K_1 - K_2}{N} + (R_2 - R_a)\frac{L_2 - K_3 - A}{N} - \frac{R_a(\sum_{i=1}^{3} K_i)}{N}, \quad (23)$$

$$y_2 = y_{02} + (R_1 - R_a)\frac{L_1 - K_1 - K_2}{N} + (R_2 - R_a)\frac{L_2 - K_3 - A}{N} - \frac{R_a(\sum_{i=1}^{3} K_i)}{N}. \quad (24)$$

There are now again seven equations which are (10), (22), (12)–(14), (23), (24) but the list of unknowns is now R_1, R_2, T_1, T_2, T_3, y_1, y_2 while exogenous are R_a, L_1, L_2, K_1, K_2, K_3, y_{01}, y_{02}, N and A. The last of these is a policy instrument the planner chooses so as to maximize the expected value of consumer welfare given by (5). We will denote the welfare maximizing value as A^*_{UGB}. This can be compared to A_{TOLLS}, the market solution for A when traffic tolls are levied and to $A_{\text{NO-TOLLS}}$ the market solution when tolls are not levied.

3. Mobile jobs and residences: no cross-commuting

In the first set of numerical examples we present, we arbitrarily assume that there is no cross-commuting. This means that we will set $\Delta_{12} = \Delta_{21} = 0 \Rightarrow P_{12} = P_{21} = 0$ (from (4)): each consumer is restricted to work in the zone in which he resides but is free to choose in which zone to reside-work. Thus (4) implies, $P_{11} > 0$, $P_{22} > 0$ and $P_{11} + P_{22} = 1$. The reason for this simplification is to make the interpretation of the results easier before the assumption is relaxed in the next subsection. NP_{ii} is the number of jobs as well as residents in zone i. The two binary logit choice probabilities are:

$$P_{11} = \frac{\exp[\lambda \ln(y_1 - T_{11}) - \lambda b \ln R_1]}{\sum_{i=1}^{2} \exp[\lambda \ln(y_i - T_{ii}) - \lambda b \ln R_i]} \quad \text{and}$$

$$P_{22} = \frac{\exp[\lambda \ln(y_2 - T_{22}) - \lambda b \ln R_2]}{\sum_{i=1}^{2} \exp[\lambda \ln(y_i - T_{ii}) - \lambda b \ln R_i]}. \quad (25)$$

As explained above, $R_2 \geqslant R_a$ depending on whether there is a binding UGB in place or not.

3.1. The reference case

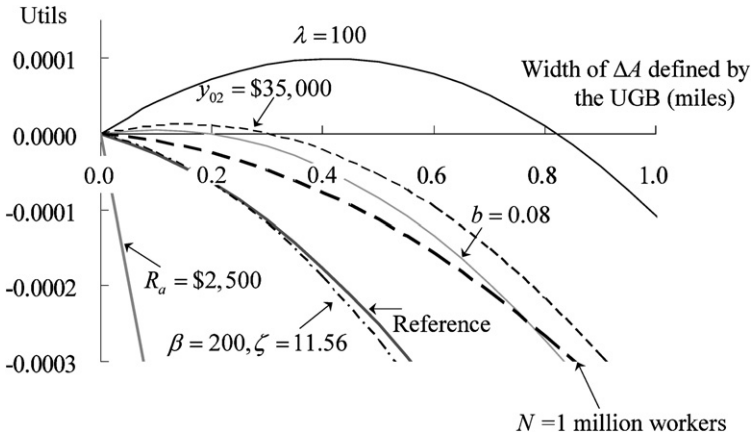
Our choice of initial parameters is as follows. We assume $a = 0.95$ and $b = 0.05$. Thus the share of land in disposable income is 5%. The taste heterogeneity parameter is set to $\lambda = 10$. As shown in Fig. 1, zone 1 has an outer radius of 3 miles and zone 2 has an inner radius of 3 miles and an outer radius of 20 miles and thus a radial length of 17 miles. Total acres in these zones are $L_1 = 18{,}095$ and $L_2 = 786{,}120$. Thus zone 1 represents a relatively small central area where congestion in real cities is typically high, while zone 2 stands for the rest of the urban area. For simplicity we will often refer to zone 1 as the city and to zone 2 as the suburbs. For the congestion functions, we set $\beta = 100$, $\zeta = 13.49$, $\gamma = 1.5$, $\alpha_1 = 20$, $\alpha_3 = 50$. Note that the parameter α_i represents the length of road i, which is important because the total cost of travel on a road is a function of its length as well as its congestion level which depends on the road's capacity. Road 1 serves the CBD while road 3, the SBD. Because there is no cross-commuting, road 2 is unused presently and its land is available for residences in zone 1 (see Fig. 1). Thus, $K_2 = 0$. Land devoted to roads 1 and 3 is set such that $K_1 = 5428$ acres (30% of L_1) and $K_3 = 43{,}428$

acres (5.52% of L_2). The agricultural land rent in zone 2 is $R_a = \$1724$ per acre. Total population is set at $N = 500{,}000$ consumer/workers for the entire circular urban area. The earned incomes are set at $y_{01} = y_{02} = \$32{,}000$ per annum. We call this choice of parameters the reference case.

Before tolls are levied, about 78% of the population resides and works in zone 2 with the remaining 22% residing and working in zone 1. Thus, although the urban area is not monocentric, it retains a somewhat centralized appearance with many jobs in a small central core of a 3 mile radius. The transport cost of city residents is about 11.4% of their incomes and that of suburban residents is about 8.7%. This is because the city congestion level on road 1 is 2.2 times higher than the suburban congestion level on road 3, as measured by $F_1/(\zeta K_1)$ relative to $F_3/(\zeta K_3)$, which dominates the fact that, the suburban road is longer as implied by $\alpha_3 > \alpha_1$. Rent per acre in the city is 7 times higher than in the suburbs and gross densities are similarly 7.3 times higher. The rent dividend is only \$98 per year because the aggregate net rent collected from the residential part of zone 1 goes almost entirely to rent from the farmers, the land needed for roads 1 and 3.

Seven numerical examples are presented in Fig. 3 and Tables 2 and 3. Four of these examples, tabulated in Table 2, include the reference case and in each of these $A^*_{\text{UGB}} = A_{\text{NO-TOLLS}}$, and therefore, a restrictive UGB which is a second best policy to congestion tolls, does not exist. In these cases, the existence of an expansive (or negative) UGB is implied. In the three examples tabulated in Table 3 a restrictive UGB that improves welfare does exist ($A^*_{\text{UGB}} > A_{\text{NO-TOLLS}}$) but the restrictiveness of such a UGB as measured by the width of the urban land annulus it transfers to agriculture is limited to 0.414 miles in one case, to about 0.101 miles in another and to 0.15 miles in the third. In other words, the urban radius shrinks by at most 2.8% when an optimal second-best UGB does exist.

We now need to understand what causes the dichotomy that separates the cases of Table 2 in which a restrictive UGB does not exist as a second-best and the cases of Table 3 in which a restrictive UGB does exist as a second best policy to congestion tolls. In order to do this, it is important to discern the direct and indirect effects of tolls.



Note. Values of vertical axis are welfare of a UGB minus welfare of no-UGB, no-tolls.

Fig. 3. UGB and welfare under no cross commutes.

Table 2
Congestion tolls expand the urban area and any restrictive UGB is inefficient

	No cross-commuting between city and suburbs							
	(1) Reference case		(2) More congestion		(3) More consumers		(4) Higher farming rent	
	see text		$\beta: 100 \to 200$ $\zeta: 13.49 \to 11.56$		$N: 0.5 \to 1$ million		$R_a: \$1724 \to \$2500$	
	Market	Tolls	Market	Tolls	Market	Tolls	Market	Tolls
P_{11}	0.218	0.204	0.204	0.195	0.195	0.191	0.228	0.210
P_{22}	0.782	0.796	0.796	0.805	0.805	0.809	0.772	0.790
T_{11} (\$/yr)	3644	8251	8326	19,408	8693	21,025	3902	8586
T_{22} (\$/yr)	2778	7048	7103	17,986	8108	20,349	2725	6975
t_1 (\$/yr)	–	4938	–	11,633	–	12,603	–	5140
t_3 (\$/yr)	–	4199	–	10,762	–	12,180	–	4155
R_1 (\$/ac/yr)	12,225	11,354	9559	9069	18,009	17,574	12,643	11,489
R_2 (\$/ac/yr)	1724	1724	1724	1724	1724	1724	2500	2500
Div. (\$/yr)	98	4425	30	10,949	122	12,377	13	4345
$y_1 - T_{11}$	28,453	28,175	23,704	23,541	23,429	23,352	28,111	27,759
$y_2 - T_{22}$	29,319	29,377	24,927	24,963	24,014	24,027	29,288	29,370
$(y_2 - T_{22}) - (y_1 - T_{11})$	866	1202	1223	1422	585	675	1177	1611
$F_1/(\zeta K_1)$	1.486	1.394	1.628	1.555	2.659	2.604	1.556	1.432
$F_3/(\zeta K_3)$	0.668	0.679	0.792	0.801	1.375	1.382	0.659	0.675
$\frac{F_1/(\zeta K_1)}{F_3/(\zeta K_3)}$	2.22	2.05	2.05	1.94	1.93	1.88	2.36	2.12
$\sum_{i=1,3} F_i g_i$ (bill. \$/year)	1.48334	1.47182	3.67629	3.66597	8.22226	8.21781	1.49656	1.47571
Welfare	10.1547	10.1550	9.9907	9.9910	9.95217	9.95224	10.1363	10.1368
$\%\Delta W$		+0.003		+0.003		+0.0007		+0.005
Radius (miles)	14.000	14.113	13.177	13.247	17.592	17.637	11.9615	12.0850
ΔRadius		+0.113		+0.070		+0.045		+0.1235

Note. No UGB columns appear because there is no second-best optimal restrictive UGB.

3.2. Direct and indirect effects of tolls

Levying tolls on the reference situation has direct and indirect effects. Figure 2 diagrams set the context for understanding these effects. To understand first the direct effect of tolls, assume initially that the unit rents in the two zones do not change when the tolls are levied. Then, the tolls will cause population (jobs and residences) to shift from zone 1 to 2 if disposable income, $y - T = y_0 + D - g - t$, increases in zone 2 relative to zone 1. Since the dividend, D,[4] which includes the average revenue from tolls does not depend on location, disposable income in zone 2

[4] Recall that D includes redistributed toll revenue and differential rents and is seen on the right sides of Eqs. (15) and (16) in the market city, (20) and (21) in the tolled city, and (23) and (24) in the city constrained by a UGB.

Table 3
Tolls centralize the urban area and some restrictive UGB is a second-best policy

	No cross-commuting between city and suburbs								
	(5) Higher suburban income			(6) Higher income share for land			(7) Less taste variety		
	y_{02} : \$32,000 → \$35,000			b : 0.05 → 0.08			λ : 10 → 100		
	Market	Tolls	Second-best UGB	Market	Tolls	Second-best UGB	Market	Tolls	Second-best UGB
P_{11}	0.1648	0.1741	0.1654	0.169	0.176	0.170	0.127	0.158	0.130
P_{22}	0.8352	0.8259	0.8346	0.831	0.824	0.830	0.873	0.842	0.870
T_{11} (\$/yr)	2407	6502	2421	2500	6617	2511	1640	5613	1687
T_{22} (\$/yr)	3059	7448	3056	3036	7420	3034	3265	7668	3251
t_1 (\$/yr)	–	3889	–	–	3958	–	–	3356	–
t_3 (\$/yr)	–	4439	–	–	4422	–	–	4571	–
R_1 (\$/ac/yr)	9636	10,272	9679	15,849	16,654	15,903	7620	9593	7778
R_2 (\$/ac/yr)	1724	1724	1766	1724	1724	1747	1724	1724	1847
Div. (\$/yr)	32	4391	64	189	4550	216	−19	4410	68
$y_1 - T_{11}$	29,625	29,890	29,643	29,689	29,933	29,705	30,341	30,797	30,381
$y_2 - T_{22}$	31,973	31,943	32,008	29,153	29,130	29,182	28,716	28,742	28,817
$(y_2 - T_{22}) - (y_1 - T_{11})$	2348	2054	2366	−536	−803	−523	−1625	−2055	−1564
$F_1/(\zeta K_1)$	1.125	1.189	1.130	1.154	1.203	1.158	0.869	1.078	0.886
$F_3/(\zeta K_3)$	0.713	0.705	0.712	0.709	0.703	0.709	0.745	0.719	0.743
$\frac{F_1/(\zeta K_1)}{F_3/(\zeta K_3)}$	1.578	1.687	1.587	1.628	1.711	1.633	1.166	1.499	1.192
$\sum_{i=1,3} F_i g_i$ (bill. \$/year)	1.47597	1.47006	1.47547	1.47290	1.46916	1.47260	1.52893	1.48232	1.52408
Welfare	10.2348	10.2350	10.2348	10.0883	10.0884	10.0883	10.1107	10.112	10.1108
$\%\Delta W$		+0.002	+0.0001		+0.0008	+0.000001		+0.012	+0.001
Radius (miles)	14.939	14.861	14.789	17.610	17.535	17.509	14.537	14.323	14.123
ΔRadius		−0.078	−0.150		−0.075	−0.101		−0.214	−0.414

will improve relative to zone 1, provided that the congestion externality and therefore the toll is sufficiently higher in zone 1. More precisely, population and the demand for land shift from zone 1 to zone 2 if the difference $(y_2 - T_3) - (y_1 - T_1) = (t_1 + g_1) - (t_3 + g_3) - (y_{01} - y_{02})$ increases when tolls are levied. In other words, $[(t_1 + g_1) - (t_3 + g_3)]|_{\text{after-tolls}} > [g_1 - g_3]|_{\text{before-tolls}}$, namely the incurred travel cost difference between zone 1 and zone 2 increases after tolls are levied. This expresses the condition that as long as congestion and therefore the congestion toll on the city-road (road 1) is sufficiently high relative to the congestion (and the toll) on the suburban road (road 3), then the imposition of tolls will increase the disposable income of locating in the suburbs relative to that of locating in the city and tolls will induce population and jobs to shift from zone 1 to zone 2. In our reference case in Table 2, the disposable income advantage of zone 2 over zone 1 improves from \$866 before tolls are levied to \$1202 after tolls are levied.

We can now turn to the second, indirect, effect of tolls. As disposable income in zone 2 improves according to the direct effect, jobs and residences will shift from zone 1 to zone 2.

This shift will cause the demand for land in zone 1 to be lowered and rent in zone 1 to decrease. Meanwhile in zone 2, arbitrage will keep rents anchored at the agricultural rent, R_a. As a result, the higher demand for residing in zone 2 will be accommodated by the market taking land away from agriculture and putting it into urban use, thus causing the urban radius to expand.

The lowered rent in zone 1 induces a substitution effect in favor of residing and working in zone 1. However, if this drop in rent is not sufficiently large, this substitution effect will not be able to dominate the direct effect of tolls which arises from the change in disposable income, and population will shift from the city to the suburbs. The ratio of zone 2 to zone 1 population is NP_{22}/NP_{11}. The causes of an increase in P_{22}/P_{11} can be seen directly by dividing one choice probability given by (4), by the other. Doing this gives:

$$\ln\left(\frac{P_{22}}{P_{11}}\right) = \lambda \ln\left(\frac{y_2 - T_{22}}{y_1 - T_{11}}\right) - \lambda b \ln\left(\frac{R_a}{R_1}\right); \quad P_{11} + P_{22} = 1. \tag{26}$$

This shows how the log of the population ratio of the two zones is determined entirely by the composition of the two effects of tolls. The first, direct effect is measured by $\ln(\frac{y_2 - T_{22}}{y_1 - T_{11}})$ and is weighted by λ, while the second indirect effect is measured by $\ln(\frac{R_a}{R_1})$ and has a smaller weight λb. A second equation below, (27), is obtained by dividing the suburban land market clearing equation with the city's land market clearing equation. It shows that the aggregate demand for suburban land, $NP_{22}q_{22}$, where q_{22} is demanded lot size, increases relative to the demand for city land, $NP_{11}q_{11}$, in lockstep with the direct and indirect effects discussed above.

$$\ln\left(\frac{P_{22}q_{22}}{P_{11}q_{11}}\right) = (\lambda + 1)\ln\left(\frac{y_2 - T_{22}}{y_1 - T_{11}}\right) - (\lambda b + 1)\ln\left(\frac{R_a}{R_1}\right) = \ln\frac{L_2 - K_3 - A}{L_1 - K_1 - K_2}. \tag{27}$$

Solving this, and recalling that $K_2 = 0$ for the case of no cross-commuting, we get the market-determined agricultural area, A, explicitly:

$$A = L_2 - K_3 - (L_1 - K_1)\left(\frac{y_2 - T_{22}}{y_1 - T_{11}}\right)^{1+\lambda}\left(\frac{R_a}{R_1}\right)^{-(1+\lambda b)}. \tag{28}$$

We can see from (28) that as the direct effect measured by $(\frac{y_2 - T_{22}}{y_1 - T_{11}})^{1+\lambda}$ is increased by the tolls, A decreases while the indirect effect measured by $(\frac{R_a}{R_1})^{-(1+\lambda b)}$ works to offset this decrease in A.

Table 2 shows the effect of changing selected parameters of the reference case one at a time. The common characteristic of the four examples of Table 2 is that tolls expand the aggregate urban land use and thus the urban radius. In particular, the examples show that: *Tolls expand the urban area in the reference case and also when there is more congestion, a higher population, or a higher agricultural rent, than in the reference case*.

In these examples, the parameters are such that congestion, measured by $F/\zeta K$, is around twice as high in zone 1 (road 1) than it is in zone 2 (road 3). Therefore, imposing tolls on all roads will raise the disposable income position of zone 2 residents relative to zone 1 residents as shown in Table 2 by $(y_2 - T_{22}) - (y_1 - T_{11})$. This income effect of tolls induces people to suburbanize to zone 2. Although tolls also cause rent to decrease in zone 1, the net effect is that the direct income effect of tolls dominates the indirect substitution effect caused by the rent decrease in zone 1. In this way, tolls contribute to the amelioration of total travel cost by causing resident-workers to locate in the suburbs. This causes the urban radius to expand while the aggregate transport cost decreases (shown by $\sum_{s=1,3} F_s g_s$ in Table 2). In contrast, the UGB of any restrictiveness would drive people from the less congested zone 2 to the more congested zone 1. So the two policies (tolls and the UGB) act in a completely opposite way and a UGB of

any stringency is not a second best policy. Because the UGB is not a second-best policy, there are no UGB columns shown in Table 2.

Table 3 shows three additional numerical examples chosen to illustrate the second-best optimality of UGBs. The common characteristic of these examples, as opposed to those of Table 2, is that tolls contract the aggregate urban land use. The reason is that under the parameter values of these examples, tolls confer a higher disposable income advantage to residing and working in zone 1 relative to zone 2 as shown in Table 3 by $(y_2 - T_{22}) - (y_1 - T_{11})$: *Tolls contract the urban area when suburban wage is higher, the demand for land is higher, or there is less taste dispersion than in the reference case*. (Recall that the reference case is shown in Table 2.)

In each of these examples, the parameter changes decrease congestion in zone 1 and increase it in zone 2, relative to the reference case (which is in Table 2). Therefore, in Table 3, congestion in zone 1 is not as high relative to congestion in zone 2 as in the examples of Table 2. Although zone 1 is still more congested than zone 2, the tolls per consumer in zone 2 are higher than the corresponding tolls in zone 1, because of the longer trips (longer road) in zone 2. This means that tolls increase the disposable income of city resident-workers relative to that of suburban resident-workers, opposite of the direct effect in the cases of Table 2. Therefore, population and land demand shift in favor of the city and the indirect effect of tolls is to increase city rents and densities. Since the demand for suburban land has decreased but the rent in zone 2 cannot fall due to arbitrage, more land reverts to agriculture. The direct income effect favors zone 1 while the indirect rent effect favors zone 2. The direct income effect dominates the indirect rent effect. This drives people to zone 1 which decreases aggregate transport cost (shown by $\sum_{s=1,3} F_s g_s$ in Table 3) while the urban radius shrinks with the imposition of tolls. In this case, the UGB (like tolls) pushes people inward and there is a second-best efficient UGB. This suggests that the UGB will work as a second-best policy when the suburbs are sufficiently congested. This resembles, perhaps, what has happened in the southern part of Seoul, where new towns were set up recently without the adequate provision of infrastructure like roads and mass transit. Importantly, the examples of Table 3 do not resemble the stylized facts of the monocentric city in which a second-best UGB always exists. In the stylized monocentric city, the center is always much more congested than is the fringe. In Table 3 the center is more congested than the suburbs but not as much as is typically assumed in the standard model.

An extended observation as to why the congestion-increasing effects of the examples of Table 2 and those of Table 3 result in diametrically opposed conclusions about the welfare implications of the UGB can be given as follows. As we saw, the congestion-increasing effects of the examples of Table 2 (an across-the-board increase in congestion, a higher population, a higher agricultural rent) make any UGB inefficient, while the congestion-increasing effects of the examples of Table 3 (higher suburban incomes, higher demand for land, less idiosyncratic taste dispersion) cause a second-best beneficial UGB to exist. An important difference is that the congestion-increasing effects in Table 2 are not discriminatory between the zones in a major way, but those in Table 3 are. Consider the factors of the examples of Table 3 relative to the reference case which is shown in Table 2. In Table 3, higher suburban income than in the reference case, induces more people to move to zone 2 from zone 1, less idiosyncratic taste heterogeneity likewise causes people to move to zone 2 because they become more sensitive to land rent and are therefore attracted to the cheaper land in zone 2, and a higher demand for land acts likewise. As a result, in all of these cases of Table 3, the ratio of congestion in zone 1 relative to that in zone 2 decreases a lot from 2.2 in the reference case (first column in Table 2) to about 1.5 or lower. In the case of the examples of Table 2, however, the factors are not as directly discriminatory between zones. For example, a higher population or a higher agricultural rent cause congestion

in both zones to increase more or less apace, while a generalized increase in road congestion acts in a similar way. Therefore, the ratio of city to suburban congestion in Table 2 does not change a lot from the reference case, remaining around 2.

Figure 3 illustrates the relationship between welfare and the second-best UGB stringency for each of the seven examples of Tables 2 and 3, all without cross commuting, presented in this section. In the figure, the cases that were tabulated in Table 2 correspond to curves that do not peak, showing that a restrictive UGB of any stringency decreases welfare. The cases of Table 3 correspond to the inverse U-shaped curves and in each of these three cases, the optimal UGB is of a very slight stringency requiring a reduction in urban radius by about 0.1 to 0.4 miles. Notably, however, the reduction in urban radius caused by the congestion tolls is even slighter. The reason is that, as explained in Anas and Rhee [2], since the UGB does not change the marginal cost of travel, it is a blunt instrument for reducing travel cost.

It is now useful to gather what we have learned into the following summary.

When is a UGB of any stringency not a second-best policy under no cross-commuting? For the setup of Fig. 1, when centers are highly congested relative to the suburbs, then first-best efficiency requires higher congestion tolls to be levied on central traffic than on suburban traffic. In the absence of cross-commuting between the suburbs and the center, some consumer-workers react to the higher central tolls by moving their job and residence from the center to the suburbs. This increases the demand for land in the suburbs and the urbanized area expands, which is efficient, because together with expansion comes lower aggregate commuting cost and higher utility. An urban growth boundary of any stringency is not a second-best efficient policy because it restricts the urban radius, not letting it expand.

Remark 1. When tolls on traffic serve to expand the urban area, lowering aggregate commuting costs but increasing geographic sprawl, then the planners should not use the restrictive UGB. Instead, planners should set a rural boundary beyond the rural-urban fringe, preventing the agricultural use of land within this boundary and enabling the urban use of land in this area at below agricultural rents. This policy would then be a second-best policy to congestion tolls. In the context of Fig. 3, the curves that do not peak in the positive quadrant would be peaking in the negative quadrant if they were allowed to continue in that direction. We have not bothered to calculate such "negative" urban boundaries (what might be called "expansive growth boundaries"), since to our knowledge they are not real policy prescriptions anywhere in the world, though perhaps they should be.

4. Mobile jobs and residences: cross-commuting

Modern cities exhibit a high degree of cross-commuting (refer again to Table 1). A large percentage of suburbanites commute to the city centers and significant numbers also commute from the city centers out to the suburbs. In the previous section, we ruled out cross-commuting in order to simplify the interpretation and presentation of the results as much as possible. We now allow cross-commuting, setting $\Delta_{ij} = 1,\ \forall(i, j)$, so that all four probabilities, $P_{11}, P_{22}, P_{12}, P_{21}$ of the discrete choice model are positive. How would the model and the results change with such cross-commuting?

Note first that road 2 of Fig. 1 would come into use. Suburban residents working at the CBD and city residents working at the SBD would share this road causing congestion to occur on it. Second, commuters now have additional margins of adjustment as compared to the case where cross-commuting was not allowed. For example, suburban residents commuting to the CBD can

adjust to tolls or to the boundary as they would in a monocentric city, by relocating their residence to the city and thus cutting the length of their commute, economizing in this way on travel outlays. Instead of commuting over roads 3, 2 and 1, they would commute over road 1 only. While this is the only adjustment they could make in the monocentric city model, it is not the only adjustment they could make in the current model. Some will adjust to tolls by switching to a suburban job reducing their commutes to using only road 3. Some others will adjust by flipping job and residence location (residing in the city and reverse commuting to the suburbs) using only road 2. Finally, those who like their existing commuting arrangement a great deal, will adjust by doing nothing and keeping their existing commuting arrangement. Which adjustment a particular commuter will make depends on the draw of their idiosyncratic tastes, after tolls are instituted or the boundary is installed. Hence, the distribution of commuters over the four commuting arrangements depends on the distribution of the idiosyncratic tastes in the population of commuters.

Table 4 shows just three examples all involving cross commuting. The first example is the new reference case. All parameters are the same as in the reference case of Table 2, except that α_1, the coefficient of road 1, has been reduced from 20 to 13, while that of road 2 has been introduced with $\alpha_2 = 10$. The road land allocations are adjusted so that roads 1 and 2 take up 26.7 and 35.6% respectively of the land in zone 1, while road 3 still takes up 5.52% of the land in zone 2. Before tolls are introduced, about 60% of commutes are suburb-to-suburb, 17% are city-to-suburb, about 12.5% are city-to-city and only about 10% are suburb-to-city.[5] Note from the reference case of Table 4 that the introduction of tolls has the biggest impact on those with the longest commutes, namely on suburban residents working at the CBD. Tolls cause their percentage to fall dramatically from about 10 to about 2% while the percentages of all the other commutes increase. Note also that tolls reduce the level of congestion on each of the three roads. Tolls shrink the urban area by about 0.4 miles. A second-best optimal UGB exists in this case and works by shrinking the urban radius by another 0.112 miles beyond the shrinkage achieved by tolls. Note, however, that this optimal UGB has a much smaller impact on suburb-to-city commuters than do tolls. Under the UGB, the percentage of suburb-to-city commuters is reduced slightly from 10 to only 9.95% (rather than to the 2% under the tolls). Also, the optimal UGB increases congestion slightly on roads 1 and 2, reducing congestion very slightly on road 3. Because the optimal UGB fails so badly in reducing suburb-to-city commuting, it increases welfare only by a tiny amount, a little more than a mere 2% of the welfare gain achieved by tolls.

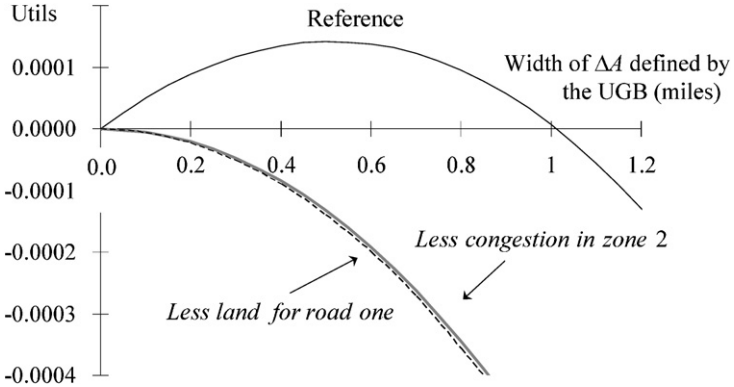
Example 2 in Table 4, perturbs the above reference case by somewhat increasing the cost of free-flow travel on road 2 and reducing it a lot on road 3. We do this by increasing α_2 from 10 to 13 and reducing α_3 from 50 to 20. These changes have the effect of causing more residents as well as more jobs to move to the suburbs. Tolls imposed on this situation, again greatly reduce the suburban residents commuting to the city, this time from 11.57 to 4.33%. And tolls again reduce congestion on each road. Tolls shrink the urban radius by a tiny 0.027 miles. In this situation any UGB, however slight in stringency, is welfare reducing. This example is interesting because unlike the examples of Section 3, the UGB fails as a second-best policy, even though tolls shrink the urban radius. The reason the UGB fails is that even though it shrinks the urban area, it has asymmetric effects on the different commuter types as we already saw in the reference case

[5] These splits compare well with the empirical ones of Table 1 recalling that our zone 1 is quite small compared to most central cities implied in Table 1. As explained earlier, that is intentional because we want to represent the central congested core (which is a small part of the central city) versus the rest of the urban area.

Table 4
Under cross-commuting and higher city congestion, any UGB is inefficient

	Cross-commuting between city and suburbs						
	(1) Reference			(2) Less suburban congestion		(3) Less city road capacity	
	Changes from ref. of Table 2 $\alpha_1 : 20 \to 13$; $\alpha_2 : 0 \to 10$; $K_1 : 0.300L_1 \to 0.267L_1$; $K_2 = 0.0L_1 \to 0.356L_1$			Changes from (1) $\alpha_2 : 10 \to 13$; $\alpha_3 : 50 \to 20$		Change from (2) $K_1 : 0.267L_1 \to 0.219L_1$	
	Market	Tolls	Second-best UGB	Market	Tolls	Market	Tolls
P_{11}	0.1225	0.1497	0.1246	0.1007	0.1102	0.0933	0.1006
P_{12}	0.1699	0.2096	0.1727	0.1256	0.1382	0.1365	0.1476
P_{21}	0.1006	0.0204	0.0995	0.1157	0.0433	0.1016	0.0350
P_{22}	0.6069	0.6203	0.6033	0.6579	0.7083	0.6686	0.7168
T_{11} (\$/yr)	2931	4869	2948	2799	4175	3221	4670
T_{12} (\$/yr)	1956	3824	1973	2144	3488	2101	3520
T_{21} (\$/yr)	7402	14,053	7412	6091	10,382	6462	10,909
T_{22} (\$/yr)	2515	5361	2490	1147	2719	1140	2720
t_1 (\$/yr)	–	2914	–	–	2497	–	2794
t_2 (\$/yr)	–	2288	–	–	2085	–	2104
t_3 (\$/yr)	–	3186	–	–	1619	–	1620
R_1 (\$/ac/yr)	31,940	40,944	32,552	24,584	27,580	22,051	24,295
R_2 (\$/ac/yr)	1724	1724	1891	1724	1724	1724	1724
Div. (\$/yr)	229	3416	328	129	2149	132	2148
$y_1 - T_{11}$	29,299	30,547	29,380	29,329	39,974	28,912	29,479
$y_1 - T_{12}$	30,273	31,592	30,355	29,985	30,660	30,031	30,629
$y_2 - T_{21}$	24,827	21,363	24,916	26,038	23,766	25,670	23,239
$y_2 - T_{22}$	29,714	30,055	29,837	30,981	31,430	30,992	31,428
$F_1/(\zeta K_1)$	1.714	1.307	1.721	1.662	1.179	1.826	1.271
$F_2/(\zeta K_2)$	1.559	1.325	1.568	1.390	1.046	1.372	1.052
$F_3/(\zeta K_3)$	0.624	0.565	0.620	0.682	0.663	0.679	0.663
$\sum_{i=1,2,3} F_i g_i$ (bill. \$/year)	1.48157	1.03943	1.47386	1.00546	0.6693	1.00308	0.66998
Welfare	10.1934	10.2027	10.1936	10.2271	10.2341	10.2259	10.2329
$\%\Delta W$		0.091%	0.001%		0.068%		0.069%
Radius (mi.)	13.340	12.941	12.829	14.093	14.066	14.076	14.079
ΔRadius		−0.399	−0.511		−0.027		+0.002

of this section. Like tolls, the UGB reduces suburb-to-city commuting which is in the efficient direction, but it also reduces suburb-to-suburb commuting which is in the inefficient direction, since tolls increase suburb-to-suburb commuting. Now in example 2 of Table 4, suburb-to-suburb commuters are more numerous than they are in the reference case of Table 4. By reducing suburb-



Note. Values of vertical axis are welfare of a UGB minus welfare of no-UGB and no-tolls.

Fig. 4. UGB and welfare under cross commutes.

to-suburb commuters, the UGB hurts more than it helps by only slightly reducing suburb-to-city commuters. As the UGB reduces suburb-to-suburb commuting it induces an underutilization of the ample suburban road capacity. The UGB also pushes people to the very congested zone 1 without providing a remedy that ameliorates the central congestion. This increases congestion on roads 1 and 2, counter to the efficient effects of the tolls policy which reduces congestion on roads 1 and 2.

Example 3 of Table 4 further perturbs example 2 by reducing the capacity of road 1 by 18%, thus making the center even more congested. Because of this, tolls now cause more shifting to the suburbs reducing aggregate travel cost but expanding the urban area and creating more geographic sprawl. Then, a UGB of any stringency is now more welfare reducing than is a UGB of the same stringency in the context of Example 2 of Table 4. The relationships between welfare and UGB stringency for the examples of Table 4 with cross-commuting are shown in Fig. 4.

The above results can be summarized as follows:

When is a UGB, of any stringency, not a second-best policy under cross-commuting? For the setup of Fig. 1, when cross-commuting takes place between the suburbs and the center, tolls impact most severely suburb-to-city commuters whose commutes are the longest. Tolls reduce suburb-to-city commuting and increase suburb-to-suburb commuting. The urban area expands (shrinks) if the demand for land in zone 2 by suburb-to-city commuters is reduced less (more) than the demand for land in zone 2 by suburb-to-suburb commuters is increased. The UGB has qualitatively different effects on different types of commuters. Unlike tolls, the UGB causes both suburb-to-city and suburb-to-suburb commuting to be reduced. Thus the UGB, of any stringency, can be inefficient when tolls shrink the city, if the inefficiency in the reduction of suburb-to-suburb commuters by the UGB is large enough relative to the efficiency of the reduction of suburb-to-city commuters.

Remark 2. As we observed in Remark 1, expansive growth boundaries, not the restrictive UGB are the better policy answer when the tolls expand the urban area. We now also see that, when cross commuting is possible and the UGB causes a sufficiently big reduction in suburb-to-suburb commuting, then urban expansion via an expansive growth boundary, not the restrictive UGB would improve welfare. (In Fig. 4, the curves that do not peak in the positive quadrant would peak in the negative quadrant.)

5. Expansion to the urban limit

In Table 5, we present two more examples. In these examples, the urban area expands up to the natural limit because surrounding land's agricultural rental value is too low. Under this assumption, which we also used in Anas and Rhee [2], we cannot show that the UGB is ever a second-best policy to congestion tolls.

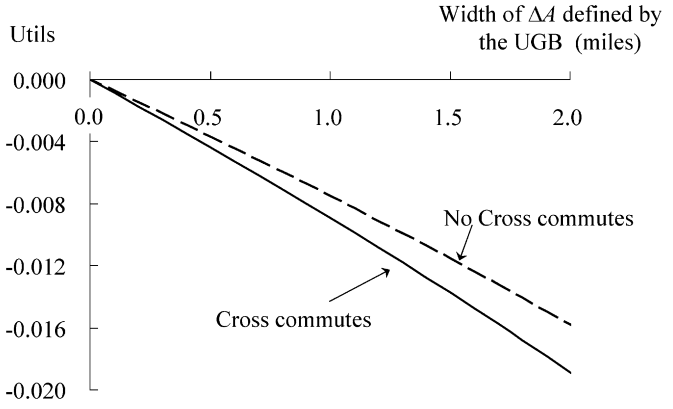
Table 5
When urban area expands to a natural limit

	(1) Without cross commuting		(2) With cross commuting	
	$a_1 = 20$; $a_3 = 50$; $K_1 = 0.300L_1$; $K_2 = 0.0L_1$; $K_3 = 0.116L_2$		$a_1 = a_2 = 13$; $a_3 = 50$; $K_1 = 0.267L_1$; $K_2 = 0.356L_1$; $K_3 = 0.112L_2$	
	Market	Tolls	Market	Tolls
P_{11}	0.1320	0.1541	0.0518	0.0912
P_{12}	0	0	0.1451	0.1768
P_{21}	0	0	0.0523	0.0081
P_{22}	0.8680	0.8459	0.7508	0.7239
T_{11} (\$/yr)	1731	5415	943	2179
T_{12} (\$/yr)	–	–	1590	3588
T_{21} (\$/yr)	–	–	5564	12303
T_{22} (\$/yr)	3238	7719	3031	6536
t_1 (\$/yr)	–	3237	–	1300
t_2 (\$/yr)	–	–	–	2145
t_3 (\$/yr)	–	4601	–	3891
R_1 (\$/ac/yr)	10,093	12,039	30,097	42,711
R_2 (\$/ac/yr)	2645	2571	2427	2223
Div. (\$/yr)	2015	6406	2032	5443
$y_1 - T_{11}$	32,283	32,990	33,090	35,264
$y_1 - T_{12}$	–	–	32,442	33,855
$y_2 - T_{21}$	–	–	31,469	28,140
$y_2 - T_{22}$	33,776	33,687	34,001	33,908
$F_1/(\zeta K_1)$	0.901	1.052	0.800	0.763
$F_2/(\zeta K_2)$	–	–	1.137	1.066
$F_3/(\zeta K_3)$	0.741	0.722	0.708	0.646
$\sum_{i=1,2,3} F_i g_i$ (bill. \$/year)	1.51973	1.48640	1.42313	1.14492
Welfare	10.2355	10.2363	10.2570	10.2619
$\%\Delta W$		+0.0076%		+0.0481%
Radius (mi)	14	14	14	14

Note. $b = 0.06$; $\lambda = 15$; $\beta = 100$; $\gamma = 1.5$; $\zeta = 13.49$; $N = 0.5$ mill; $y_{01} = \$32{,}000$; $y_{02} = \$35{,}000$; width of zone 2 = 11 miles.

To generate these examples a zero agricultural rent was used, and we assumed that there is a natural limit at 14 miles from the center to which the urban area expanded. From the diagrams in Fig. 2, we can see that when the demand for land in zone 2 is high enough, then the demand function intersects the supply of land to zone 2 high on its vertical portion which is what happens when the urban area expands to its natural limit.[6] In this situation, imposing tolls would either increase densities in zone 1, decreasing them in zone 2 or act in the opposite way increasing densities in zone 2 and decreasing them in zone 1, but without any change in the urban radius, provided the demand for land in zone 2 continues to remain sufficiently high. But the UGB would always shrink the urban radius.

The first example in Table 5 is obtained by modifying case 5 of Section 3 (Table 3) in which we did not allow cross-commuting and suburban earned income was set at $y_{02} = \$35{,}000$. We also increased the value of the taste heterogeneity parameter, λ, from 10 to 15 and the income share of land, b, from 0.05 to 0.06. These parameters were intentionally selected from the examples presented earlier in which the UGB was a second-best policy. However, parameter values make no difference in the margin as long as the equilibrium is occurring well on the vertical portion of supply. The second case is identical to the first except that cross-commuting is allowed. In both cases, welfare drops rapidly with the stringency of a UGB as seen in Fig. 5. The rapid drop is due to two facts. One of these is that tolls lower the demand curve for land in zone 2 but not sufficiently to shrink the city whereas the UGB shrinks the city. The other fact comes from having assumed a zero agricultural rent. This means that nothing is saved in land acquisition costs (the opportunity cost of urban land) when the urban radius is limited by the UGB. These two facts, natural boundary and zero agricultural rent, were also present in Anas and Rhee [2] but not in Arnott [4], Brueckner [6], Kanemoto [8], or Pines and Sadka [11] and our examples presented there showed that the UGB was inefficient even in the monocentric setup.



Note. Values of vertical axis are welfare of a UGB minus welfare of no-UGB, no-tolls.

Fig. 5. UGB and welfare with no opportunity cost of land.

[6] In the case where land is worthless in any other use, suppose that the demand curve for land in zone 2 cuts the horizontal axis. This would mean a zero land rent in zone 2 and consequently a demand for infinite lot sizes. This would drive up rents in zone 2 and the equilibrium intersection of demand and supply would have to occur in the vertical part of the zone 2 supply curve.

6. The UGB: a second-best or a wrong bet?

We showed that the nearly 30 years old intuition about the UGB being a second-best policy in congested urban areas, derived from monocentric models, clouds our understanding of the issues. Modern cities are characterized by jobs that can be relocated and are not pinned in the CBD, by cross-commuting and by the importance of heterogeneity in the choice of alternative job and residential arrangements. Once these important aspects of modern cities are recognized and suitable theoretical tools are used to model them as we have done here, then it becomes clear that a UGB policy is not always a second-best policy. In fact, more often than not, tolls would cause economic activity to shift from the congested centers to the periphery and this sets up the preconditions under which the UGB cannot be a second-best policy because it restricts the urban radius.

Indeed, in practice, it is hard to levy tolls everywhere. It is much more likely, like the recent implementation of tolls in central London, that tolls would be levied on the heavily congested central area only, that looks like our zone 1. In these situations, the shadow rent on land at the urban fringe may well exceed the agricultural rent and, in the absence of tolls, what is actually needed is an expansive growth boundary not a restrictive growth boundary (see Remarks 1 and 2 in Sections 3 and 4). In other words, contrary to their intuition, *the planners should allow the urban area to sprawl more geographically (longer radius and more land) in order to reduce aggregate travel cost (less economic sprawl)*. We also saw, in our numerical examples, that when the restrictive UGB is in fact a second-best policy, it is not a very effective second-best policy because it captures a very small (tiny) percentage of the already small welfare gains that come from tolls.

Considering planners' inability to calculate a second-best efficient UGB, in those cases where one may exist, the only slight effectiveness of the policy becomes extremely daunting. As urban areas grow and sprawl, it would appear that the UGB or similar policies become more popular. Our analysis shows that quite the opposite should happen if planners and people were better informed: in larger cities, centers are more congested and, according to our analysis, the optimal policy of congestion tolls would expand, not shrink the suburban areas. Therefore, for those properly informed by our analysis, the UGB should be a less popular policy instrument for large cities.

Acknowledgments

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